Financial-market discourse on Bluesky

Who drives the conversation, how it clusters, and how it feels (13 Aug – 12 Sep 2026)

Ege Aydemir · Web and Social Media Search and Analysis · A.Y. 2025/26 · Project type 1: Social Media Analysis

Objective

What phenomenon, for whom, with which questions

- Actor: a market-intelligence / social-listening team of a brokerage — which accounts and communities to monitor, how much to trust the platform's mood
- Topic: English financial-market discourse on Bluesky, three streams: equities · macro/Fed · crypto
- Node = user, edge = reply (directed, weighted). Post = opinion quintuple (entity, polarity, emotion, author, time)
- RQ1 structure (small-world? hubs?) · RQ2 hubs vs brokers · RQ3 communities (stable? = streams?)
- RQ4 sentiment / emotions / entities · RQ5 communities × content, echo chambers · RQ6 co-movement with SPY and BTC (exploratory)

Data: collection

Official AT Protocol API, exhaustive closed window, threads reconstructed

- 7 queries chosen after a 3-day pilot of 12 candidates (posts/day, reply share, posts/author, samples)
  - kept: "stock market", "S&P 500" · "the Fed", Warsh, FOMC · bitcoin, ethereum — all lang:en
  - dropped: "Wall Street" (9/11 posts), Nasdaq (4% replies, one bot = 41%), crypto (spam), Powell (no longer chair)
- searchPosts, sort=latest, pages of 100, 0.6 s pause, 429 handled via ratelimit-reset; window 13 Aug–13 Sep UTC (until exclusive)
- 36,322 rows → 34,533 unique posts (crypto 17.5k / macro 8.7k / equities 8.5k)
- 8,908 posts with replies → getPostThread(depth 8) → 28,556 replies; 0 failed threads
- 60,104 posts with text, 19,844 authors; median 151 characters

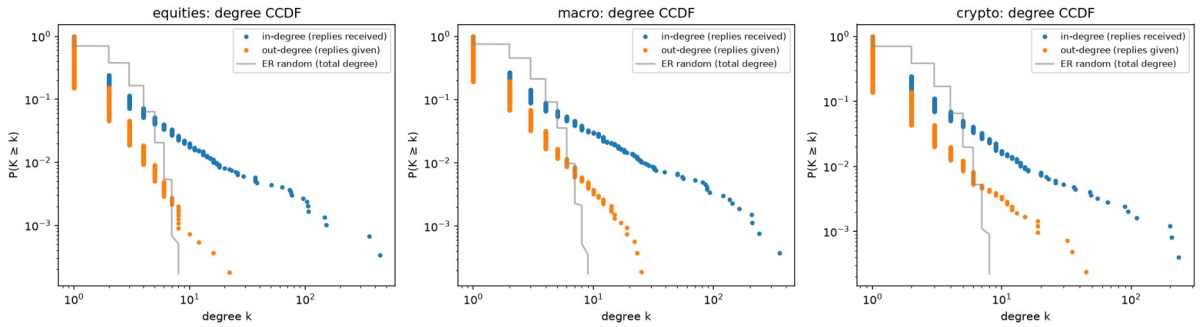
Data quality: automation, coordination, ambiguity

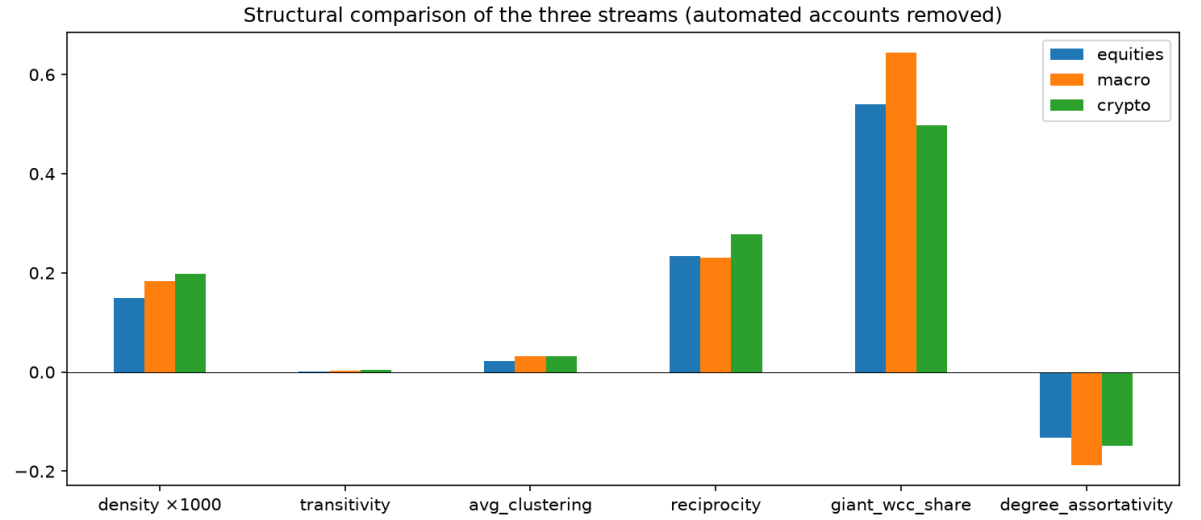
Three problems a keyword search does not tell you about

- 4 transparent rules for automated / broadcast accounts (sensitivity reported): ≥15 posts/day · ≥50% template text · broadcast-only · ≥50% near-duplicates → 73 accounts, 18.6% of posts
- Coordinated campaign: "🚨 BTC is knocking on the $80K door…" posted 965 times by 248 accounts (~10 each) — invisible to per-account rules → post-level duplicate filter (5,981 posts)
- Query ambiguity: "the Fed" = WWE in wrestling slang → a whole (fake) community; found by validating communities with content; 816 posts + their threads removed
- Removing automation changes the network by only 215 edges: automation inflates volume, not interaction
- Final: 15,973 users, 18,980 reply ties; text corpus 44,958 posts (24,589 query-matched + 20,369 replies)

RQ1 — Structure: a disassortative small world of stars

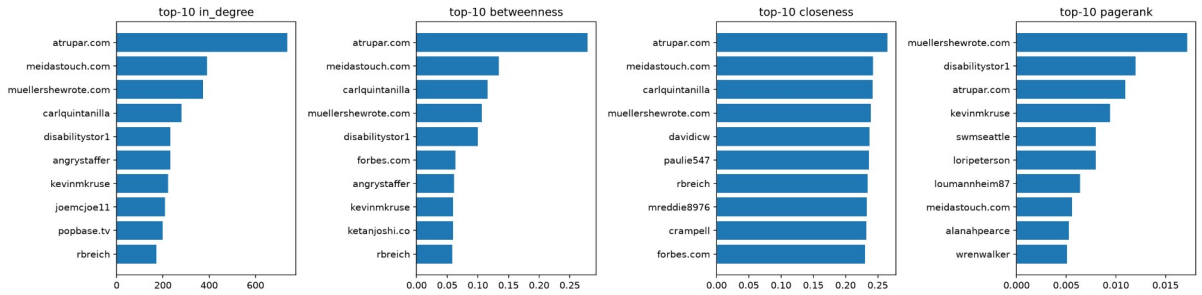
- Density ~1e-4, reciprocity 0.25, giant WCC 70% but SCC 3.6% → many users reply to few accounts that rarely reply back
- Giant component vs Erdős–Rényi G(n,m): L = 6.25 vs 9.98, C = 0.033 vs 0.0004 → Watts–Strogatz signature
- Avg clustering 0.029 ≫ transitivity 0.0023: hub effect (k(k−1)/2 weighting)
- Heavy-tailed degrees, max in-degree 738; degree assortativity −0.12 … −0.19 in every stream
- Streams: macro most connected (giant 64%), crypto deepest threads (diameter ≥ 24), equities most fragmented





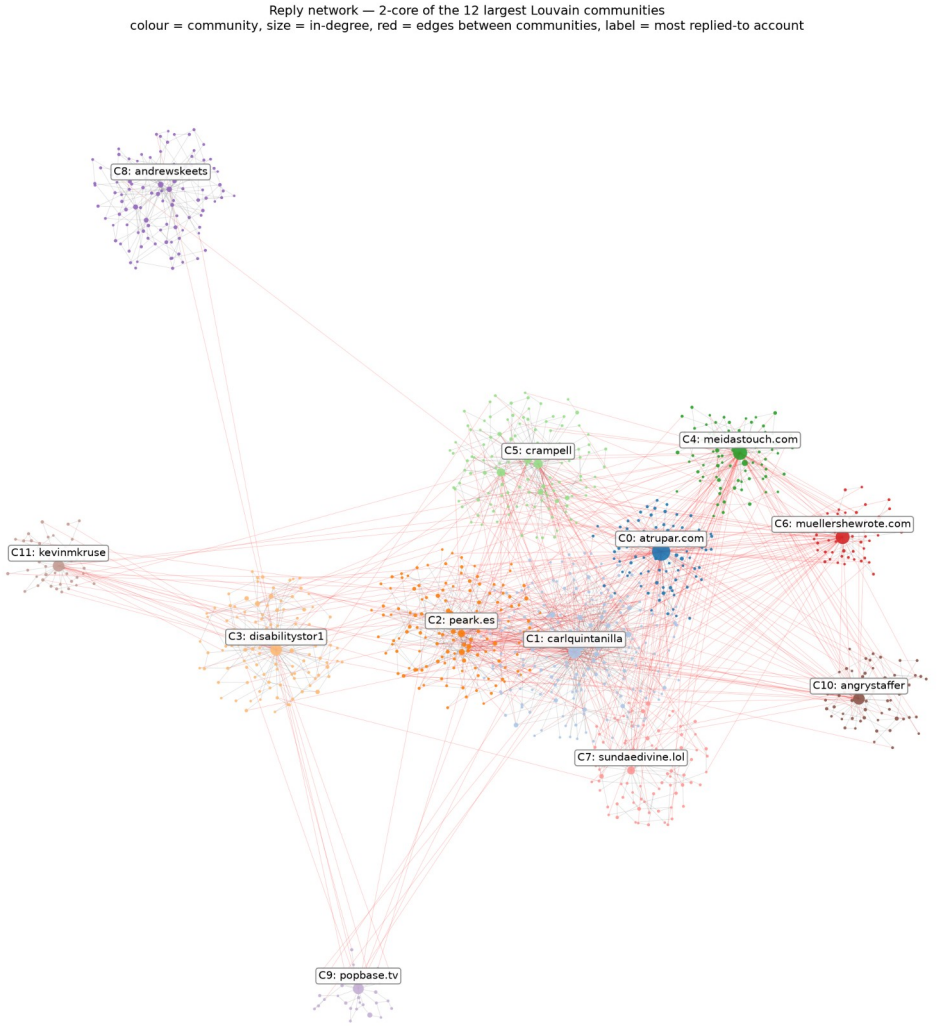
RQ2 — Hubs and brokers

- Most replied-to: atrupar.com (738), meidastouch.com (391), muellershewrote.com (373), carlquintanilla (280, CNBC) … → market talk is embedded in the political conversation
- Crypto has its own hubs: disabilitystor1, kevinmkruse, popbase.tv, ketanjoshi
- In-degree ≈ PageRank (ρ = 0.97); betweenness only moderately related (ρ ≈ 0.55)
- Brokers with degree 2–11 in the top-50 betweenness: users who reply to two hubs and bridge two audiences (connector hubs / weak ties)
- Closeness negatively related to in-degree (ρ = −0.14): being replied to ≠ being central



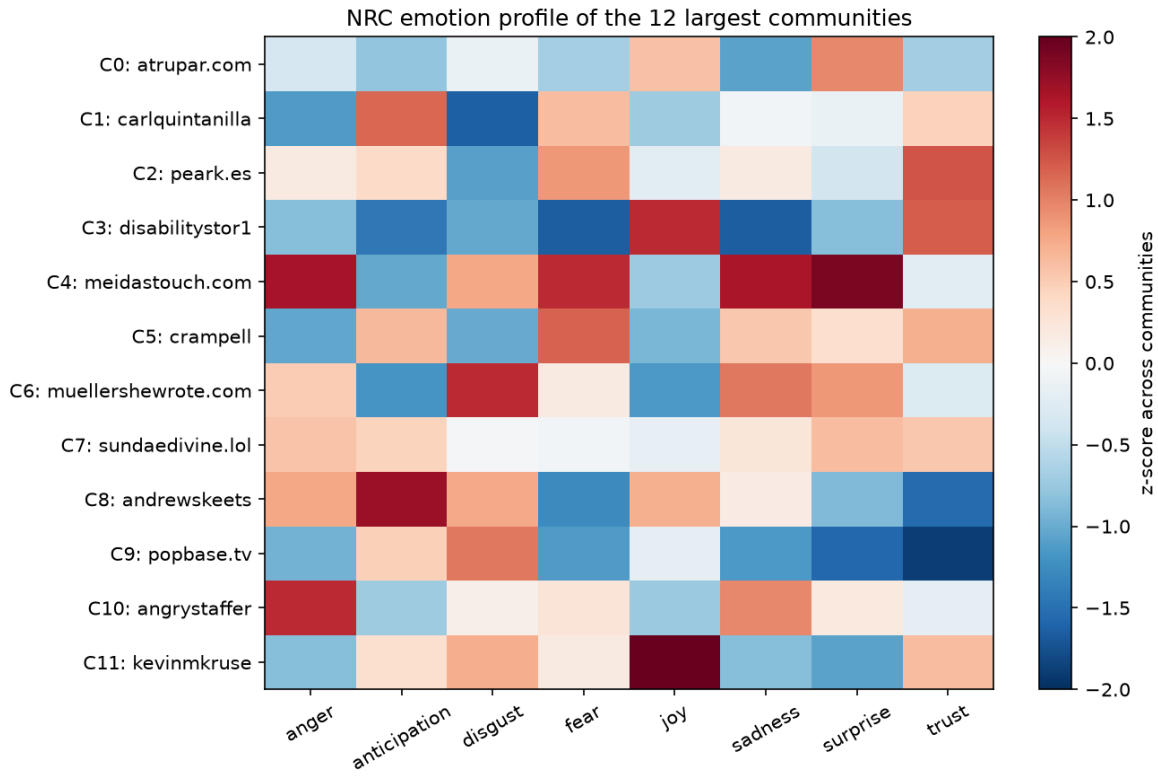
RQ3 — Communities: 67 stable star-shaped communities

- Louvain Q = 0.860 (67), CNM 0.858 (86), label propagation 0.724 (1,682), FluidC k=3 0.545 / k=10 0.748; Girvan–Newman infeasible ($O(m^2n)$)
- Stability: 5 seeds → NMI 0.90; Louvain vs CNM 0.82
- Each large community = one hub + its audience; 12.5% of edges cross communities
- Latent ≠ observable: purity vs streams 0.50–0.92; stream assortativity 0.305 (moderate homophily)
- Communities validated with content on the next slide



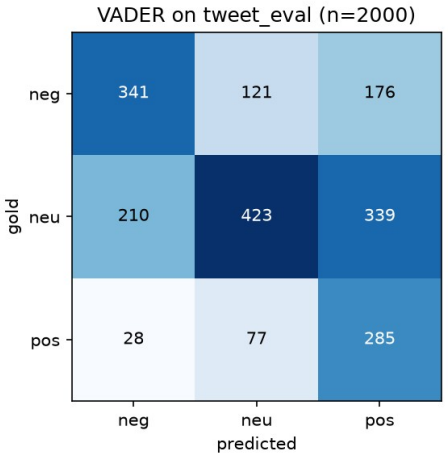
RQ5 — What the communities actually are

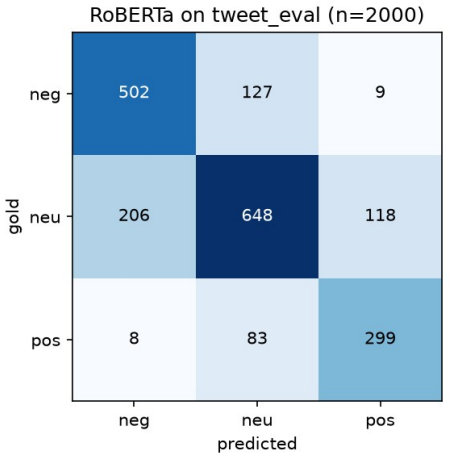
- Economics communities — carlquintanilla (Nixon, Carter, Reagan, Powell: Fed independence), peark.es / claudia-sahm (productivity, VIX, PCE, credibility): least negative (37–39%), highest trust
- Political communities — meidastouch, muellershewrote, angrystaffer, ronfilipkowski (vote, govt, Trump, job): 59–69% negative, highest anger & fear
- Off-topic threads hanging from keyword posts — popbase (a movie), an academic thread, a philosophy thread, wrestling residue
- Of the 12 largest communities ~7 are about markets → monitoring should follow the economics ones for information, the political ones for sentiment shocks



RQ4 — Sentiment models and validation

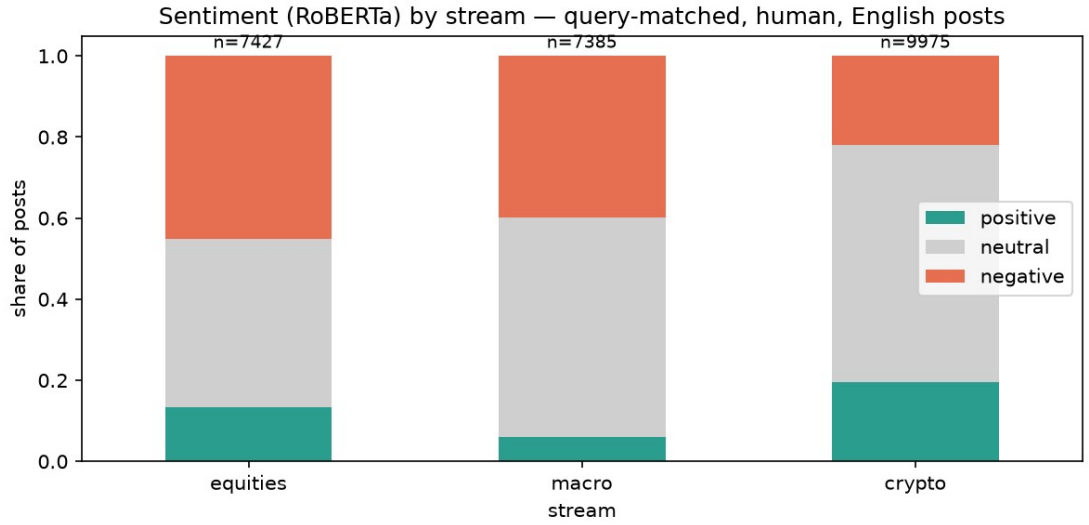
- Two families: lexicons (AFINN, VADER with negation/intensifier/emoji rules, ±0.05) and a transformer (cardiffnlp twitter-roberta, TweetEval preprocessing)
- Raw text into the models (stop-word removal would destroy negation); lemmatised text only for word statistics
- tweet_eval (2,000 labelled tweets): RoBERTa 0.724 acc / 0.728 macro-F1; AFINN 0.548; VADER 0.524
- On our posts: VADER–RoBERTa agreement 51%, κ = 0.29 — VADER calls 5,036 sarcastic political posts positive
- RoBERTa = primary model; NRC EmoLex for 8 emotions (not sentiment); no in-domain gold labels (limitation)

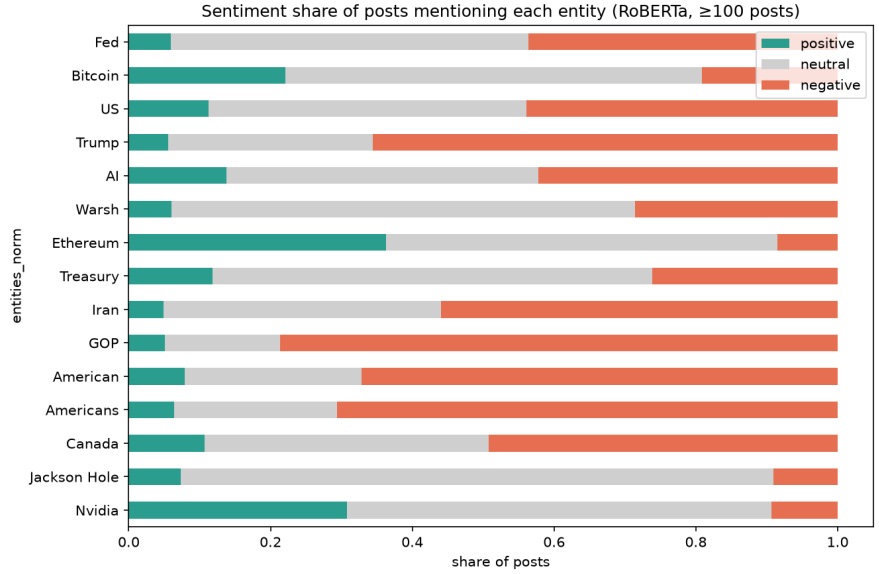




RQ4 — Streams and entities

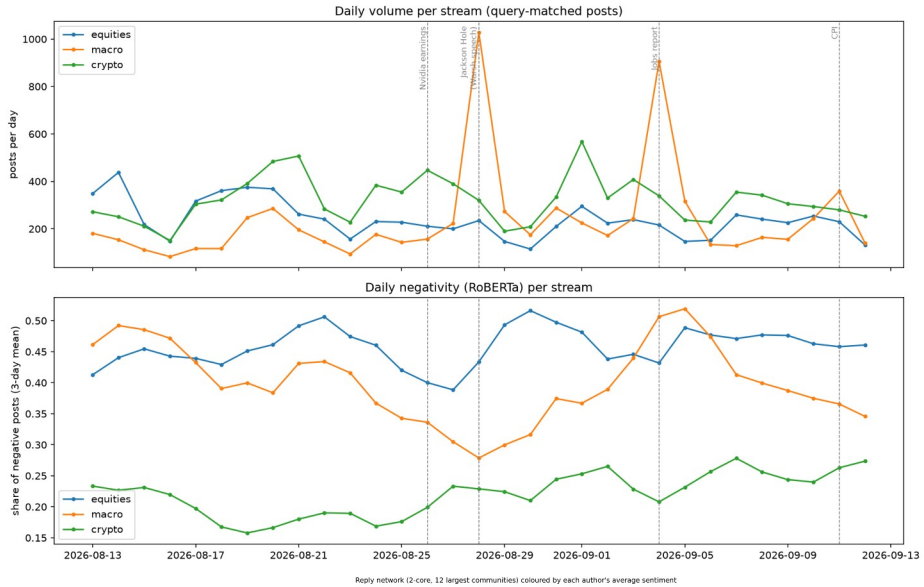
- Equities 45% negative / 13% positive; macro 54% neutral, 40% negative, 6% positive; crypto the only balanced stream (22% neg / 19% pos)
- Replies are more negative than the posts they answer (46% vs 34%)
- Entities: Fed 4,896 posts, 44% negative; Trump 66%, GOP 79%, Iran 56% negative; Bitcoin balanced; Ethereum and Nvidia the most positive; Jackson Hole 84% neutral
- Cashtags rare (799 posts): Bluesky talks markets in words, not tickers

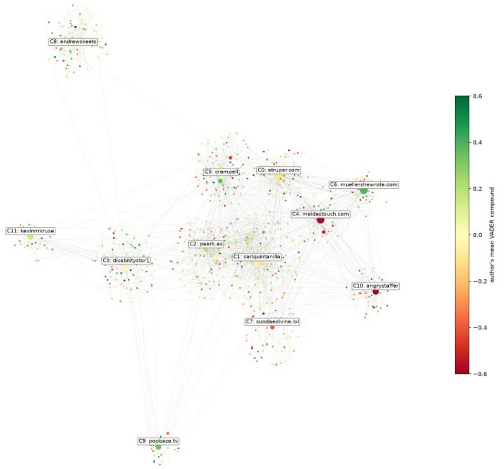




Over time, and do people reply to like-minded people?

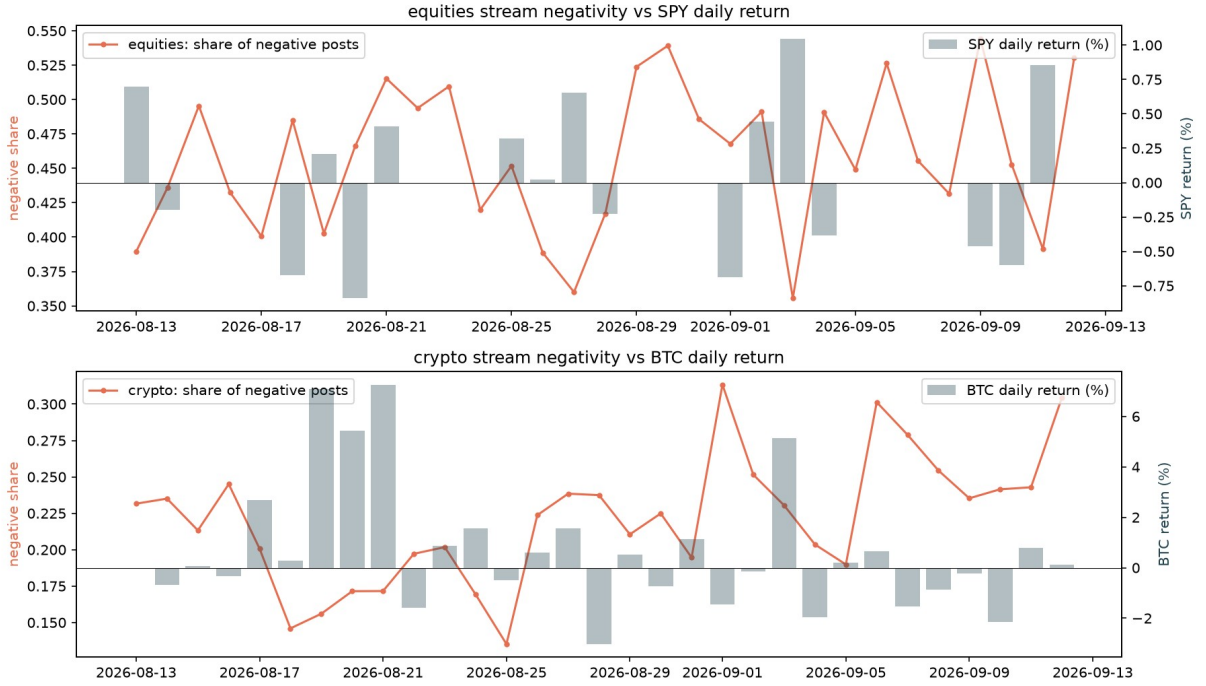
- Peaks explained by the day's distinctive terms: Unitree IPO (19 Aug), Walmart (20 Aug), Jackson Hole / Warsh (28 Aug), AI-generated Bitcoin movie (1 Sep), jobs report (4 Sep)
- Negativity in equities/macro rises around the jobs report and CPI; crypto stays least negative
- Author-level polarity assortativity 0.04 → no segregation into camps
- Reply–parent same polarity 51% vs 40% expected; negative → negative 56%, positive → positive 28% → negativity cascades down threads, positivity does not





RQ6 — Does the conversation move with the market? (exploratory)

- Daily series vs SPY (17 trading days) and BTC-USD (31 days), Pearson same day and t→t+1
- Attention ↔ activity: crypto posts vs BTC volume r = 0.70 (p < 0.001), vs |return| 0.58
- Mood ↔ same-day return: equities negativity vs SPY return r = −0.59 (p = 0.012); crypto VADER vs BTC return 0.43
- Lead–lag: weak and inconsistent; 16 tests with n = 17–31 → some significance expected by chance
- Reading: the platform reflects the market (thermometer of attention), it does not lead it — no causal claim



Limitations and knowledge for the actor

- Limitations: search returns matching posts only; large threads truncated by the API; one month; query ambiguity and thread drift handled by rules + content validation; sentiment 0.72 accuracy, sarcasm; sampled path lengths; short market series, correlation ≠ causation
- Knowledge: follow the economics communities (carlquintanilla, peark.es, claudia-sahm, crampell) for information; treat political hubs as sources of sentiment shocks; discount 1 in 5 posts as automated/coordinated; use Bluesky as a real-time thermometer of attention, not a forecasting signal
- Everything reproducible: notebooks 01–05, raw JSON Lines, fixed seeds, requirements.txt, README